

Aryan Dayal

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EDUCATION

Purdue University, School of Mechanical Engineering
Bachelor of Science in Engineering

West Lafayette, IN
Expected Graduation: May 2027
GPA: 3.8/4.0

- **Awards:** Best Research Poster (Purdue SURF 2025), Dean's List, Semester Honours

RELEVANT SKILLS

- **Mechanical Design & Prototyping:** Siemens NX, SolidWorks, GD&T, DFM, 3D Printing, CNC Machining, Prototyping
- **Testing & Experimentation:** Design of Experiments (DOE), Test Setup Design & Development, Root Cause Analysis
- **Programming & Modeling:** Python, MATLAB, Linux, Pandas, NumPy, Simulink, Data Analysis, Physics-Based Modeling

PROFESSIONAL EXPERIENCE

Purdue University – School of Mechanical Engineering

West Lafayette, IN

Undergraduate Researcher (Nanoscale Energy Transport & Conversion Lab)

January 2026 – Present

- Develop and debug **physics-based simulation workflows** using Python, Linux, LAMMPS, and Phonopy to analyze thermal transport, processing **10^3 – 10^4 atomic configurations** and resolving data inconsistencies across complex simulation pipelines
- Build automated **Python pipelines for 100+ parameter-sweep and validation runs**, reducing manual post-processing time by **~70%** and enabling repeatable testing, data analysis, and systematic troubleshooting across simulation studies
- Evaluate lattice thermal conductivity using **Boltzmann transport-based models**, troubleshooting convergence and input issues while validating outputs against physical expectations and ensuring consistency across simulation environments

Purdue University – School of Materials Engineering

West Lafayette, IN

Undergraduate Researcher (NASA-Funded | Martinez Soft Materials Group)

May 2025 – Present

- **Designed and iterated a custom monolayer deposition system** for 3 μm silica particles, refining hardware and operating parameters through 30+ experimental trials to achieve repeatable and consistent particle deposition
- **Design and troubleshoot electromechanical test setups** for electric-field-directed particle assembly in **UV-curable 3D printing materials**, integrating custom mechanical components and optical microscopy to characterize system performance
- **Plan DOE-driven validation studies (25+ test conditions)** and develop automated **Python-based data analysis** to characterize system behavior, evaluate test setup performance, and identify failure modes affecting particle alignment

Purdue University – School of Mechanical Engineering

West Lafayette, IN

Undergraduate Research Lead (Manufacturing Lab)

January 2025 – December 2025

- **Designed and iterated Gaussian-to-top-hat beam-shaping systems** for 800 nm light, translating physics-based simulation outputs into fabrication-ready geometries and debugging performance deviations to achieve **$\leq 5\%$ intensity non-uniformity**
- Applied **design of experiments (DOE)** and response optimization across **20+ design iterations**, improving transmission efficiency to **$\geq 85\%$** while satisfying manufacturing constraints and validating system performance
- **Fabricated and tested micro-scale prototypes** using two-photon lithography (TPP), SEM, and sputtering; performed measurement validation, defect analysis, and process optimization to improve structural fidelity

LEADERSHIP & INVOLVEMENT

Purdue University – School of Mechanical Engineering

West Lafayette, IN

Tutor

January 2026 – June 2026

- Provided **one-on-one tutoring** to **30+ students** in core Mechanical Engineering courses, guiding problem-solving through **free-body diagrams, equilibrium analysis, energy balances, and first-principles reasoning**

Purdue Department of Physics and Astronomy

West Lafayette, IN

Teaching Assistant: Lab

August 2024 – December 2024

- Guided **120+ students** through VPython modeling and instrumentation labs using **oscilloscopes, multimeters, Hall probes, and sensor-based measurement systems**, verifying setups and calibration for repeatable experimental data

PROJECTS

Automated Microscopy Image Analysis: Developed an automated Python-based computer vision pipeline to quantify particle assembly from microscopy images; extracted performance metrics across 25+ experimental conditions to enable data-driven system characterization, test comparison, and identification of failure modes

Electric Go-Kart Gearbox System: Designed a multi-stage gear reduction system to meet a 1,000-rpm output requirement;

performed shaft, bearing, gear-force, and stress analyses, developed the CAD assembly, and validated the design against constraints

Three-Band Audio Equalizer: Designed, built, and tested an analog signal-processing system using active filters and amplifiers; validated frequency response, cutoff accuracy, and output performance against design requirements